

Deep Learning: Theory and Practice

Neural Network Models

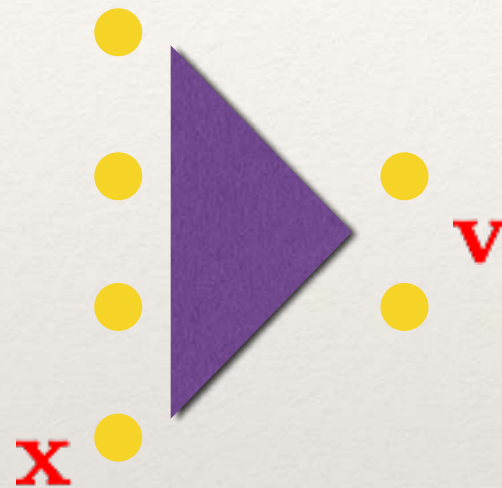
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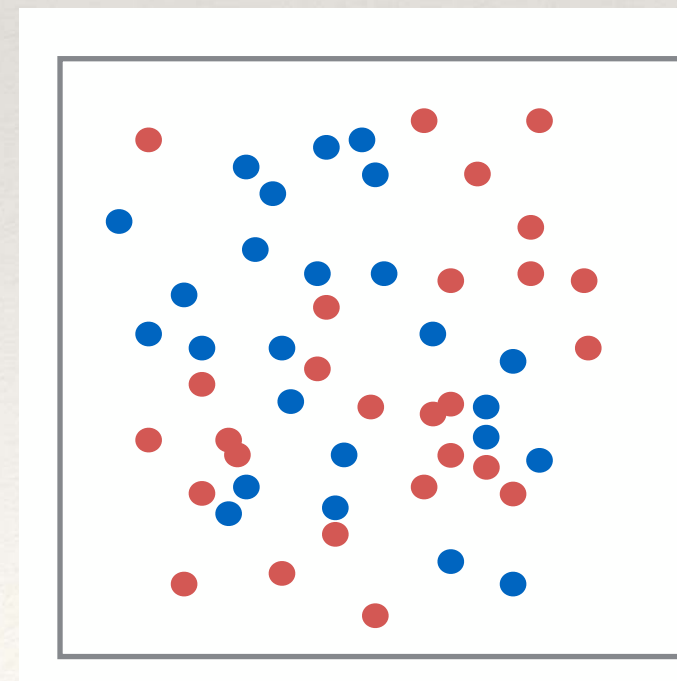
Perceptron Algorithm

Perceptron Model [McCulloch, 1943, Rosenblatt, 1957]



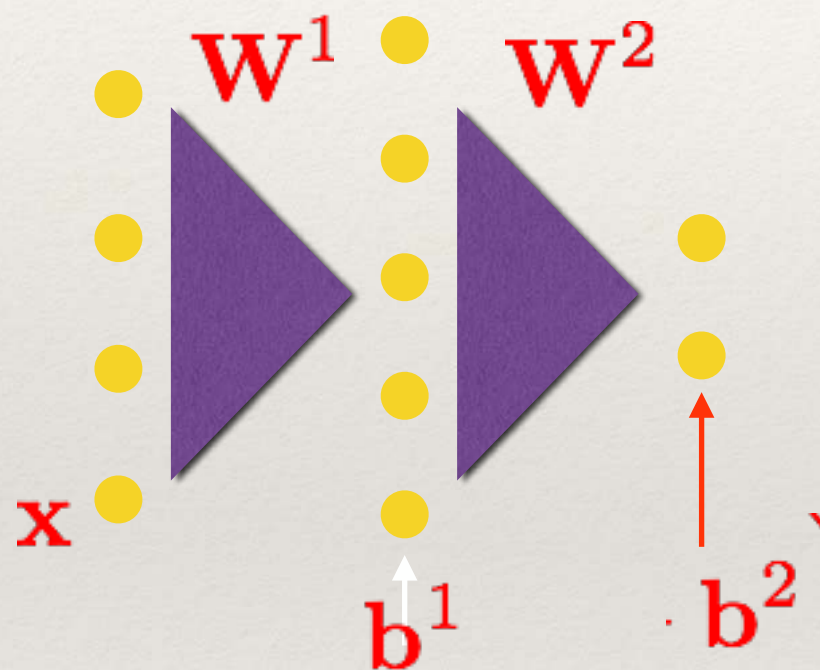
Targets are binary classes $[-1, 1]$

What if the data is not
linearly separable



Multi-layer Perceptron

Multi-layer Perceptron [Hopfield, 1982]



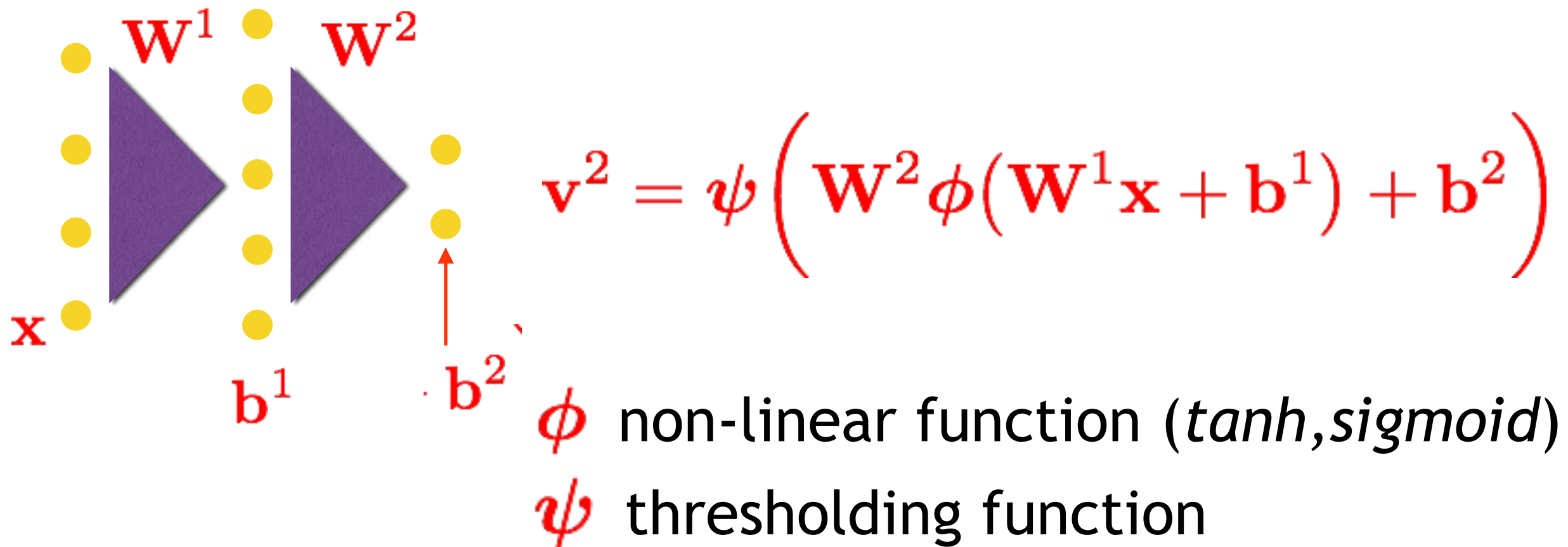
$$\mathbf{v}^2 = \psi \left(\mathbf{W}^2 \phi(\mathbf{W}^1 \mathbf{x} + \mathbf{b}^1) + \mathbf{b}^2 \right)$$

ϕ non-linear function (*tanh, sigmoid*)

ψ thresholding function

Neural Networks

Multi-layer Perceptron [Hopfield, 1982]

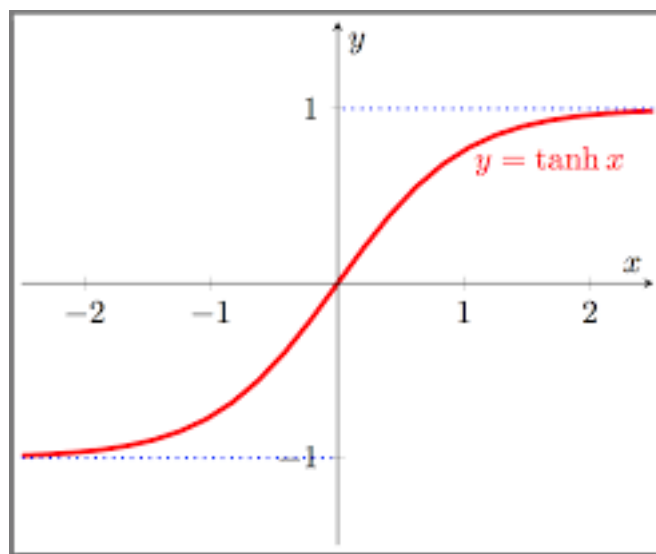


- Useful for classifying **non-linear data boundaries** - non-linear class separation can be realized given enough data.

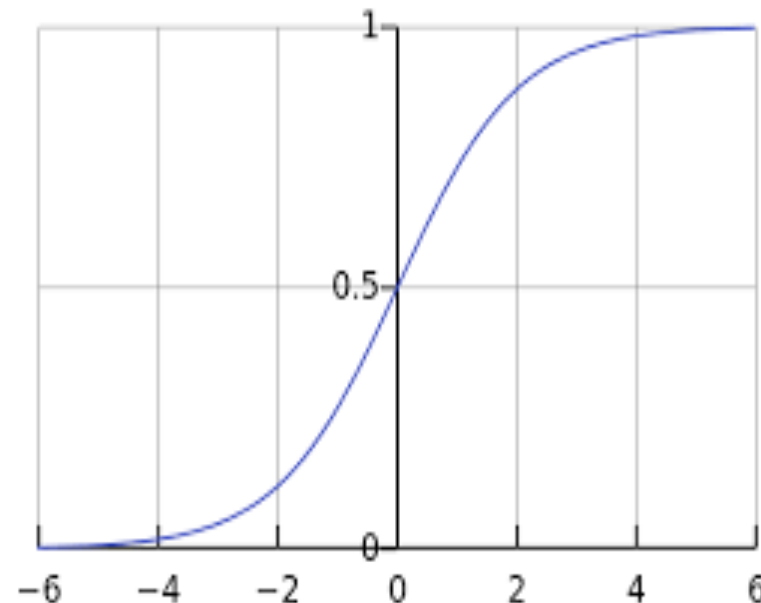
Neural Networks

Types of Non-linearities ϕ

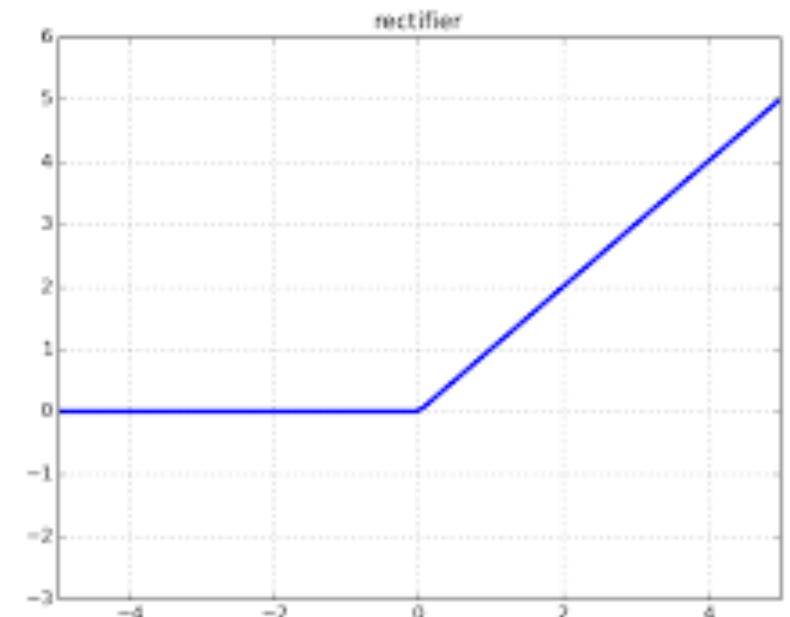
tanh



sigmoid



ReLu



Cost-Function

Mean Square Error

$$J_{MSE} = \sum_{i=1}^M ||\mathbf{v}_i - \mathbf{y}_i||^2$$

Cross Entropy

$$J_{CE} = - \sum_{i=1}^M \mathbf{y}_i^T \log(\mathbf{v}_i)$$

$\mathbf{y}_i$ are the desired outputs

Learning Posterior Probabilities with NNs

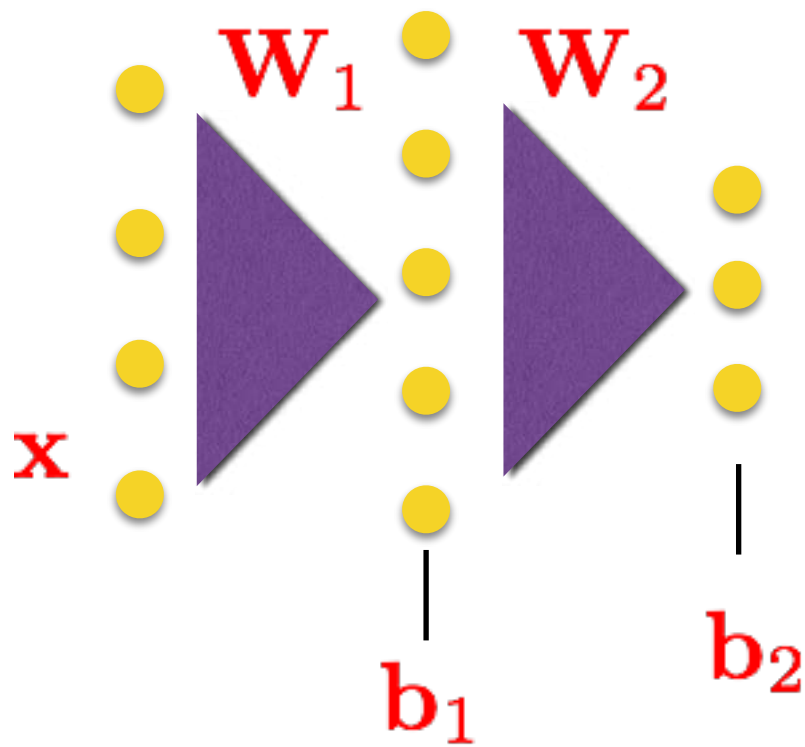
Choice of target function ψ

- Softmax function for classification

$$\psi(v_i) = \frac{e^{v_i}}{\sum_i e^{v_i}}$$

- Softmax produces positive values that sum to 1
- Allows the interpretation of outputs as posterior probabilities

Parameter Learning

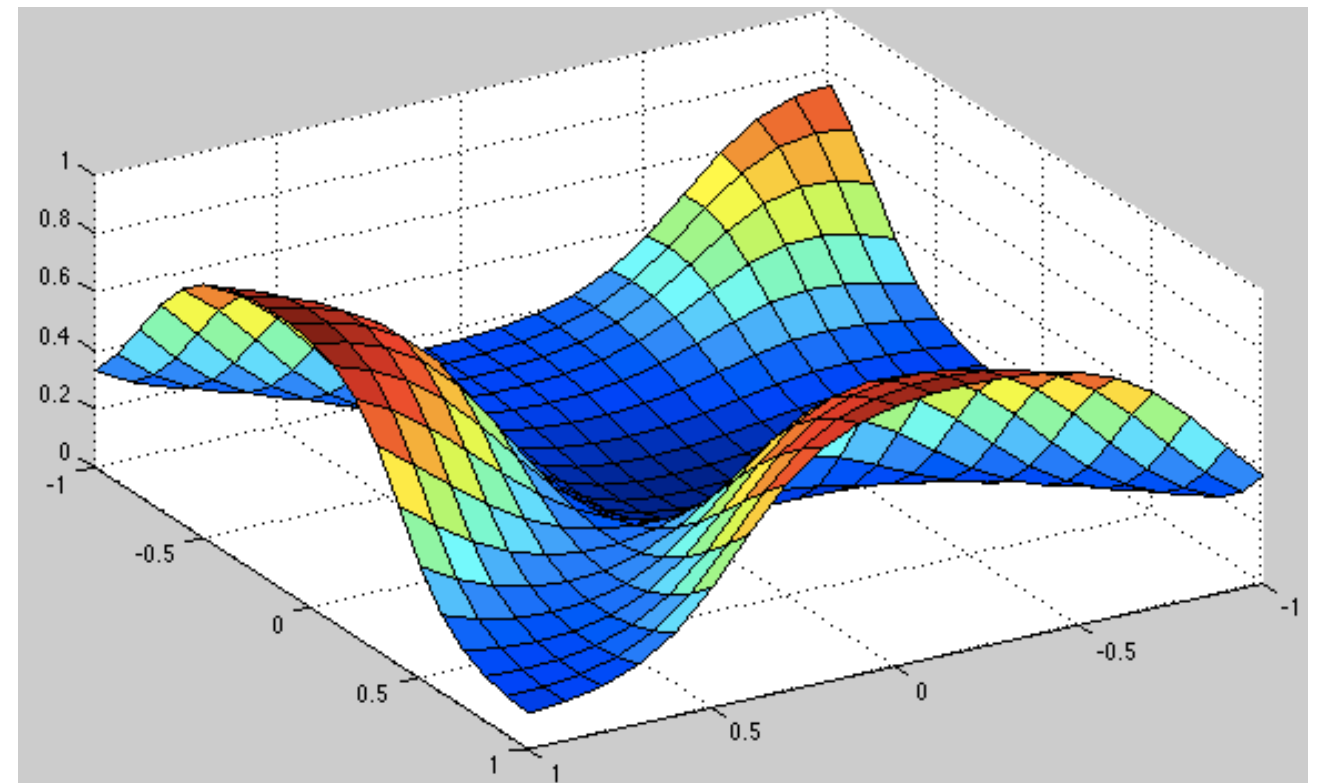


$$\mathbf{v}^2 = \psi \left(\mathbf{W}^2 \phi(\mathbf{W}^1 \mathbf{x} + \mathbf{b}^1) + \mathbf{b}^2 \right)$$

Error function for entire data

$$J_{MSE} = \sum_{i=1}^M ||\mathbf{v}_i - \mathbf{y}_i||^2$$

Typical Error Surface as a function of parameters (weights and biases)



Parameter Learning

Error surface close to a local optima

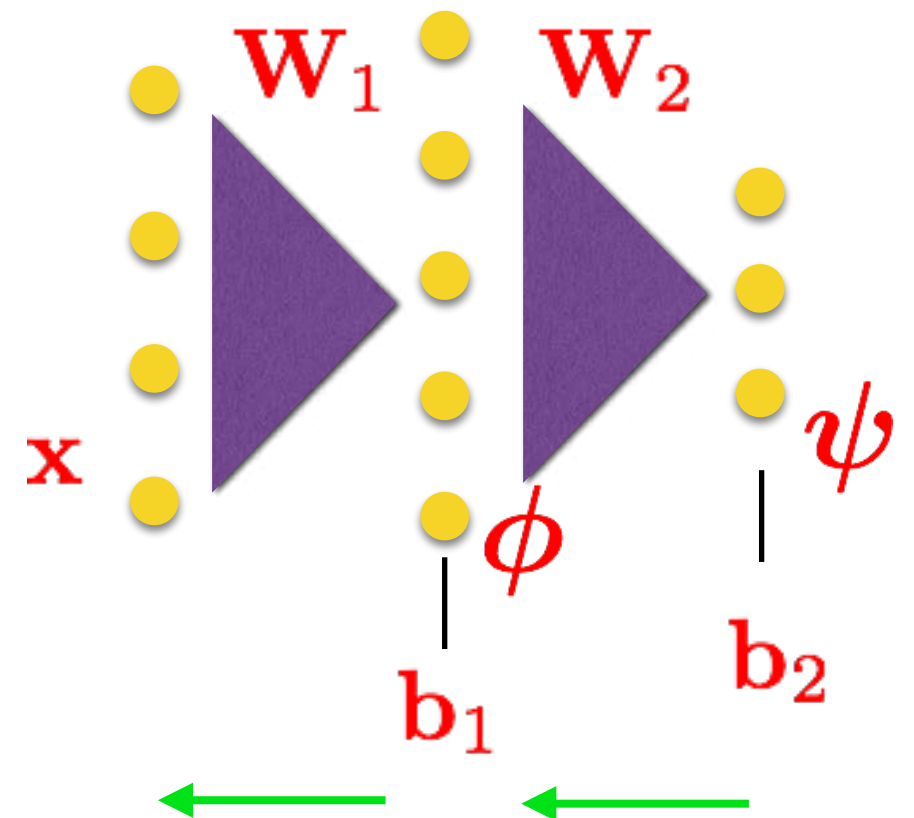
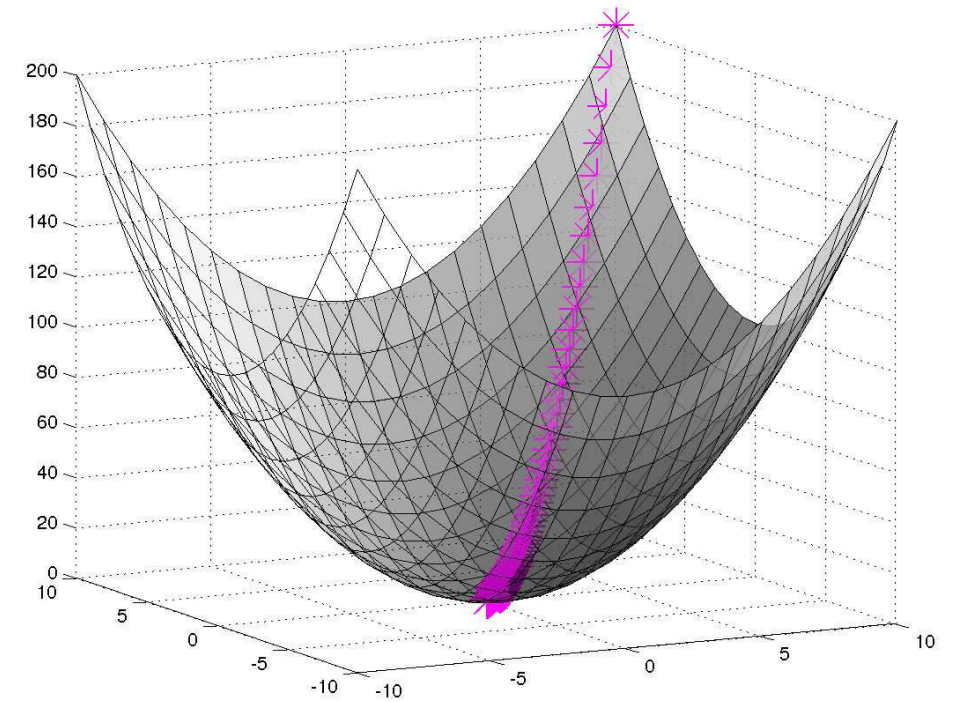
Non-linear nature of error function

- Move in the reverse direction of the gradient

$$\mathbf{W}_1^t = \mathbf{W}_1^{t-1} - \eta \frac{\partial J}{\partial \mathbf{W}_1}$$

Error back propagation

$$\frac{\partial J}{\partial \mathbf{W}_1} = \frac{\partial J}{\partial \psi} \times \frac{\partial \psi}{\partial \phi} \times \frac{\partial \phi}{\partial \mathbf{W}_1}$$



Parameter Learning

- Solving a non-convex optimization.
- Iterative solution.
- Depends on the initialization.
- Convergence to a local optima.
- Judicious choice of learning rate

